# LECTURE 07

# LIMITS OF TRIGONOMETRIC FUNCTIONS

If $\theta$ is the radian measure of a central angle of a circle of radius *r*, then the area $A$ of the sector determined by $\theta$ is

$$A = \frac{1}{2}r^2\theta$$

# LIMITS OF TRIGONOMETRIC FUNCTIONS

$$\lim_{t \to 0} sint = 0$$

$$\lim_{t \to 0} cost = 1$$

$$\lim_{t \to 0} \frac{sint}{t} = 1$$

$$\lim_{t \to 0} \frac{1 - cost}{t} = 0$$

# PROOF

$$\lim_{x\to 0}\frac{sinx}{x} = 1$$

**L.H.L:**

| x | -0.5 | -0.3 | -0.1 | -0.01 | -0.001 |
|---|---|---|---|---|---|
| f(x) | 0.9588 | 0.985 | 0.9983 | 0.99998 | 0.999999 |

$$\lim_{x\to 0^-}\frac{sinx}{x} = 1$$

# CONT…

**R.H.L:**

| x | 0.5 | 0.3 | 0.1 | 0.01 | 0.001 |
|---|---|---|---|---|---|
| f(x) | 0.9588 | 0.985 | 0.9983 | 0.99998 | 0.999999 |

$$\lim_{x \to 0^+} \frac{sinx}{x} = 1$$

**L.H.L=R.H.L**

$$\lim_{x \to 0} \frac{sinx}{x} = 1$$

# PROOF (ALTERNATIVE)

$$\lim_{x \to 0} \frac{sinx}{x}$$

**Solution:**

After putting direct limit, we have the form 0/0

So by applying L'Hôpital's rule, we get

$$\lim_{x \to 0} \frac{\frac{d}{dx}(sinx)}{\frac{d}{dx}(x)} = \lim_{x \to 0} \frac{cosx}{1} = \cos(0) = 1$$

# PROOF

$$\lim_{t \to 0} \frac{1 - cost}{t} = 0$$

**Solution:**

After putting direct limit, we have 0/0 form, so, by applying L'Hôpital's rule, we get

$$\lim_{t \to 0} \frac{\frac{d}{dt}(1 - cost)}{\frac{d}{dt}(t)} = \lim_{t \to 0} \frac{sint}{1} = \sin(0) = 0.$$

# EXAMPLE

$$\lim_{x\to 0} \frac{tanx}{x}$$

**Solution:**

Applying L'Hôpital's rule,

$$\lim_{x\to 0} \frac{\frac{d}{dx}(tanx)}{\frac{d}{dx}(x)} = \lim_{x\to 0} \frac{sec^2x}{1}$$

$$= \lim_{x\to 0} \frac{1}{cos^2x}$$

$$= \lim_{x\to 0} \frac{1}{cos^2(0)} = 1$$

# ALTERNATIVE SOLUTION

$$\lim_{x\to 0}\frac{\tan x}{x} = \lim_{x\to 0}\frac{\frac{\sin x}{\cos x}}{x}$$

$$= \lim_{x\to 0}\frac{\sin x}{x}.\frac{1}{\cos x}$$

$$= \lim_{x\to 0}\frac{\sin x}{x}.\lim_{x\to 0}\frac{1}{\cos x}$$

As we know that $\lim_{x\to 0}\frac{\sin x}{x} = 1$, so

$$= 1.\lim_{x\to 0}\frac{1}{\cos x}$$

$$= \frac{1}{\cos(0)} \Rightarrow 1$$

# EXAMPLE 1

$$\lim_{x \to 0} \frac{sin5x}{2x}$$

**Solution:**

$$\lim_{x \to 0} \frac{sin5x}{2x} = \lim_{x \to 0} \frac{1}{2} \frac{sin5x}{x}$$

$$= \lim_{x \to 0} \frac{5}{2} \frac{sin5x}{5x}$$

By using $\lim_{t \to 0} \frac{sint}{t} = 1$,

$$= \lim_{x \to 0} \frac{5}{2} \lim_{x \to 0} \frac{sin5x}{5x}$$

$$= \frac{5}{2} \lim_{x \to 0} \frac{sin5x}{5x} = \frac{5}{2}(1) = \frac{5}{2}$$

# EXAMPLE 3

$$\lim_{x\to 0}\frac{(2x+1-cosx)}{3x}$$

**Solution:**

$$\lim_{x\to 0}\frac{(2x+1-\cos x)}{3x}=\lim_{x\to 0}\left(\frac{2x}{3x}+\frac{1-cosx}{3x}\right)$$

$$=\lim_{x\to 0}\frac{2x}{3x}+\lim_{x\to 0}\frac{1-cosx}{3x}$$

By using $\lim_{t\to 0}\frac{1-cost}{t}=0$,

$$=\lim_{x\to 0}\frac{2}{3}+\lim_{x\to 0}\frac{1}{3}\left(\frac{1-cosx}{x}\right)$$

$$=\frac{2}{3}+\frac{1}{3}(0)=\frac{2}{3}$$

Find the limits in Exercises 1–26.

**1** $\lim_{x \to 0} \dfrac{x}{\sin x}$

**2** $\lim_{x \to 0} \dfrac{\sin x}{\sqrt[3]{x}}$

**3** $\lim_{t \to 0} \dfrac{\sin^3 t}{(2t)^3}$

**4** $\lim_{\theta \to 0} \dfrac{3\theta + \sin \theta}{\theta}$

**5** $\lim_{x \to 0} \dfrac{2 + \sin x}{3 + x}$

**6** $\lim_{t \to 0} \dfrac{1 - \cos 3t}{t}$

**7** $\lim_{\theta \to 0} \dfrac{2 \cos \theta - 2}{3\theta}$

**8** $\lim_{x \to 0} \dfrac{x^2 + 1}{x + \cos x}$

**9** $\displaystyle\lim_{x \to 0} \frac{\sin(-3x)}{4x}$

**10** $\displaystyle\lim_{x \to 0} \frac{x \sin x}{x^2 + 1}$

**11** $\displaystyle\lim_{x \to 0} \frac{1 - \cos x}{x^{2/3}}$

**12** $\displaystyle\lim_{x \to 0} \frac{1 - 2x^2 - 2\cos x + \cos^2 x}{x^2}$

**13** $\displaystyle\lim_{t \to 0} \frac{4t^2 + 3t \sin t}{t^2}$

**14** $\displaystyle\lim_{x \to 0} \frac{x \cos x - x^2}{2x}$

**15** $\displaystyle\lim_{t \to 0} \frac{\cos t}{1 - \sin t}$

**16** $\displaystyle\lim_{t \to 0} \frac{\sin t}{1 + \cos t}$

**17** $\lim_{t \to 0} \dfrac{1 - \cos t}{\sin t}$

**18** $\lim_{x \to 0} \dfrac{\sin (x/2)}{x}$

**19** $\lim_{x \to 0} \dfrac{x + \tan x}{\sin x}$

**20** $\lim_{t \to 0} \dfrac{\sin^2 2t}{t^2}$

**21** $\lim_{x \to 0} x \cot x$

**22** $\lim_{x \to 0} \dfrac{\csc 2x}{\cot x}$

**23** $\lim_{\alpha \to 0} \alpha^2 \csc^2 \alpha$

**24** $\lim_{x \to 0} \dfrac{\sin 3x}{\sin 5x}$

**25** $\lim_{v \to 0} \dfrac{\cos (v + \pi/2)}{v}$

**26** $\lim_{x \to 0} \dfrac{\sin^2 (x/2)}{\sin x}$

Establish the limits in Exercises 27–30, where $a$ and $b$ are any nonzero real numbers.

**27** $\lim_{x \to 0} \frac{\sin ax}{bx} = \frac{a}{b}$

**28** $\lim_{x \to 0} \frac{1 - \cos ax}{bx} = 0$

**29** $\lim_{x \to 0} \frac{\sin ax}{\sin bx} = \frac{a}{b}$

**30** $\lim_{x \to 0} \frac{\cos ax}{\cos bx} = 1$

# DERIVATIVE

Let $f$ be a function defined on an open interval containing $a$. The derivative of $f$ at $a$, written $f'(a)$, is given by

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

provided the limit exists.

**Example 1** If $f(x) = 3x^2 - 5x + 4$, find $f'(x)$. What is the domain of $f'$? Find $f'(2)$, $f'(-\sqrt{2})$, and $f'(a)$.

$$
\begin{aligned}
f'(x) &= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h \to 0} \frac{[3(x+h)^2 - 5(x+h) + 4] - (3x^2 - 5x + 4)}{h} \\
&= \lim_{h \to 0} \frac{(3x^2 + 6xh + 3h^2 - 5x - 5h + 4) - (3x^2 - 5x + 4)}{h} \\
&= \lim_{h \to 0} \frac{6xh + 3h^2 - 5h}{h} = \lim_{h \to 0} (6x + 3h - 5) \\
&= 6x - 5.
\end{aligned}
$$

Since $f'(x) = 6x - 5$ for all $x$, the domain of $f'$ is $\mathbb{R}$. Substituting for $x$, we have

$$f'(2) = 6(2) - 5 = 7$$
$$f'(-\sqrt{2}) = 6(-\sqrt{2}) - 5 = -(6\sqrt{2} + 5)$$
$$f'(a) = 6a - 5. \qquad \blacksquare$$

***Example 2*** Find $f'(x)$ if $f(x) = \sqrt{x}$. What is the domain of $f'$?

***Solution*** The domain of $f$ consists of all nonnegative real numbers. We shall examine the cases $x > 0$ and $x = 0$ separately. If $x > 0$, then by (3.6),

$$f'(x) = \lim_{h \to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}.$$

To find the limit we begin by multiplying numerator and denominator of the indicated quotient by $\sqrt{x+h}+\sqrt{x}$. Thus

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{\sqrt{x+h}-\sqrt{x}}{h} \cdot \frac{\sqrt{x+h}+\sqrt{x}}{\sqrt{x+h}+\sqrt{x}} \\ &= \lim_{h \to 0} \frac{(x+h)-x}{h(\sqrt{x+h}+\sqrt{x})} \\ &= \lim_{h \to 0} \frac{1}{\sqrt{x+h}+\sqrt{x}} \\ &= \frac{1}{\sqrt{x}+\sqrt{x}} = \frac{1}{2\sqrt{x}}. \end{aligned}$$

# CONT…

$$f'(x) = \frac{1}{2\sqrt{x}}$$

For domain $x \neq 0 \; and \; x > 0.$

So, the domain of $f'(x)$ is the set of positive real numbers.

In Exercises 1–10 use (3.6) to find $f'(x)$.

**1** $f(x) = 37$

**2** $f(x) = 17 - 6x$

**3** $f(x) = 9x - 2$

**4** $f(x) = 7x^2 - 5$

**5** $f(x) = 2 + 8x - 5x^2$

**6** $f(x) = x^3 + x$

**7** $f(x) = 1/(x - 2)$

**8** $f(x) = (1 + \sqrt{3})^2$

**9** $f(x) = \sqrt{3x + 1}$

**10** $f(x) = 1/2x$

In Exercises 11–14 find $D_x y$.

**11** $y = 7/\sqrt{x}$

**12** $y = (2x + 3)^2$

**13** $y = 2x^3 - 4x + 1$

**14** $y = x/(3x + 4)$

# RULES FOR FINDING DERIVATIVES

1) $D_x c = 0,$
2) $D_x(x) = 1,$
3) If n is a positive integer, then $D_x(x^n) = nx^{n-1},$ (The Power Rule)

***Example 1***

(a) Find $D_x(x^3)$ and $D_x(x^8)$.

(b) Find $y'$ if $y = x^{100}$.

***Solution*** Applying the Power Rule (3.13) gives us

(a) $D_x(x^3) = 3x^2$ and $D_x(x^8) = 8x^7$.

(b) $y' = D_x y = D_x(x^{100}) = 100x^{99}$.

**Rule 04:** $D_x[cf(x)] = cD_x[f(x)]$

***Example 2***

(a) Find $D_x(7x^4)$.

(b) Find $F'(z)$ if $F(z) = -3z^{15}$.

***Solution*** By the remarks preceding this example we have

(a) $D_x(7x^4) = (7)4x^3 = 28x^3$.

(b) $F'(z) = (-3)(15)z^{14} = -45z^{14}$.

**Rule 05:** $D_x[f(x) + g(x)] = D_x[f(x)] + D_x[g(x)]$
**Rule 06:** $D_x[f(x) - g(x)] = D_x[f(x)] - D_x[g(x)]$

***Example 3*** Find $f'(x)$ if $f(x) = 2x^4 - 5x^3 + x^2 - 4x + 1$.

***Solution***

$$\begin{aligned} f'(x) &= D_x(2x^4 - 5x^3 + x^2 - 4x + 1) \\ &= D_x(2x^4) - D_x(5x^3) + D_x(x^2) - D_x(4x) + D_x(1) \\ &= 8x^3 - 15x^2 + 2x - 4 \end{aligned}$$

**The Product Rule (Rule 07):** $D_x[f(x)g(x)] = f(x)D_x[g(x)] + g(x)D_x[f(x)]$

***Example 4*** Find $f'(x)$ if $f(x) = (x^3 + 1)(2x^2 + 8x - 5)$.

***Solution*** Using the Product Rule (3.16),

$$\begin{aligned} f'(x) &= (x^3 + 1)D_x(2x^2 + 8x - 5) + (2x^2 + 8x - 5)D_x(x^3 + 1) \\ &= (x^3 + 1)(4x + 8) + (2x^2 + 8x - 5)(3x^2) \\ &= 4x^4 + 8x^3 + 4x + 8 + 6x^4 + 24x^3 - 15x^2 \\ &= 10x^4 + 32x^3 - 15x^2 + 4x + 8. \end{aligned}$$

**The Quotient Rule (Rule 08)::** $D_x\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)D_x[f(x)]-f(x)D_x[g(x)]}{[g(x)]^2}$, provided $g(x) \neq 0$.

***Example 5*** Find $y'$ if $y = \dfrac{3x^2 - x + 2}{4x^2 + 5}$.

***Solution*** By the Quotient Rule (3.17),

$$\begin{aligned} y' &= \frac{(4x^2+5)D_x(3x^2-x+2)-(3x^2-x+2)D_x(4x^2+5)}{(4x^2+5)^2} \\ &= \frac{(4x^2+5)(6x-1)-(3x^2-x+2)(8x)}{(4x^2+5)^2} \\ &= \frac{(24x^3-4x^2+30x-5)-(24x^3-8x^2+16x)}{(4x^2+5)^2} \\ &= \frac{4x^2+14x-5}{(4x^2+5)^2}. \end{aligned}$$

**Rule 09:** If n is a positive integer, then $D_x(x^{-n}) = -nx^{-n-1}$

***Example 6*** Differentiate the following: (a) $g(w) = 1/w^4$. (b) $H(s) = 3/s$.

***Solution***

(a) Writing $g(w) = w^{-4}$ and using Theorem (3.18) (with $w$ as the independent variable),

$$g'(w) = D_w(w^{-4}) = -4w^{-5} = -\frac{4}{w^5}.$$

(b) Since $H(s) = 3s^{-1}$,

$$H'(s) = D_s(3s^{-1}) = 3(-1)s^{-2} = -\frac{3}{s^2}. \quad \blacksquare$$

Differentiate the functions defined in Exercises 1–32.

**1** $f(x) = 10x^2 + 9x - 4$

**2** $f(x) = 6x^3 - 5x^2 + x + 9$

**3** $f(s) = 15 - s + 4s^2 - 5s^4$

**4** $f(t) = 12 - 3t^4 + 4t^6$

**5** $g(x) = (x^3 - 7)(2x^2 + 3)$

**6** $k(x) = (2x^2 - 4x + 1)(6x - 5)$

**7** $h(r) = r^2(3r^4 - 7r + 2)$

**8** $g(s) = (s^3 - 5s + 9)(2s + 1)$

**9** $f(x) = (4x - 5)/(3x + 2)$

**10** $h(x) = (8x^2 - 6x + 11)/(x - 1)$

**11** $h(z) = (8 - z + 3z^2)/(2 - 9z)$

**12** $f(w) = 2w/(w^3 - 7)$

**13** $f(x) = 3x^3 - 2x^2 + 4x - 7$

**14** $g(z) = 5z^4 - 8z^2 + z$

**15** $F(t) = t^2 + (1/t^2)$

**16** $s(x) = 2x + 1/(2x)$

**17** $g(x) = (8x^2 - 5x)(13x^2 + 4)$

**18** $H(y) = (y^5 - 2y^3)(7y^2 + y - 8)$

**19** $G(v) = (v^3 - 1)/(v^3 + 1)$

**20** $f(t) = (8t + 15)/(t^2 - 2t + 3)$

**21** $f(x) = 1/(1 + x + x^2 + x^3)$

**22** $p(x) = 1 + (1/x) + (1/x^2) + (1/x^3)$

**23** $g(z) = z(2z^3 - 5z - 1)(6z^2 + 7)$

**24** $N(v) = 4v(v - 1)(2v - 3)$

**25** $K(s) = (3s)^{-4}$

**26** $W(s) = (3s)^4$

**27** $h(x) = (5x - 4)^2$

**28** $g(r) = (5r - 4)^{-2}$

**29** $f(t) = \dfrac{(3/5t) - 1}{(2/t^2) + 7}$

**30** $S(w) = (2w + 1)^3$

**31** $M(x) = (2x^3 - 7x^2 + 4x + 3)/x^2$

**32** $f(x) = (3x^2 - 5x + 8)/7$

In Exercises 33 and 34 find $y'$ by means of (a) the Quotient Rule (3.17), (b) the Product Rule (3.16), and (c) simplifying algebraically and using the Power Rule (3.13).

**33** $y = (3x - 1)/x^2$

**34** $y = (x^2 + 1)/x^4$

In Exercises 35 and 36 find $y'$ by (a) using the Product Rule and (b) first multiplying the two factors.

**35** $y = (12x - 17)(5x + 3)$

**36** $y = 8x^2(5x - 9)$

39 $y = (8x - 1)(x^2 + 4x + 7)(x^3 - 5)$

40 $y = (3x^4 - 10x^2 + 8)(2x^2 - 10)(6x + 7)$

41 Find an equation of the tangent line to the graph of $y = 5/(1 + x^2)$ at each point.

(a) $P(0, 5)$ (b) $P(1, \frac{5}{2})$ (c) $P(-2, 1)$

42 Find an equation of the tangent line to the graph of $y = 2x^3 + 4x^2 - 5x - 3$ at each point.

(a) $P(0, -3)$ (b) $P(-1, 4)$ (c) $P(1, -2)$

43 Find the $x$-coordinates of all points on the graph of $y = x^3 + 2x^2 - 4x + 5$ at which the tangent line is (a) horizontal; (b) parallel to the line $2y + 8x - 5 = 0$.

44 Find point $P$ on the graph of $y = x^3$ such that the tangent line at $P$ has $x$-intercept 4.